2	**(a)**	The area under F_E represents the work done by the driving force of the car's engine. [A1]	
	(b)	**(i)**	Net work done = area bounded by F_E and F_R $= \frac{1}{2}(50)(4000) = 1.0 \times 10^5$ J [C1] This is also the gain in the kinetic energy of the car. Since the car started from rest, the maximum speed of the car is: $\frac{1}{2}(1800)v^2 = 100000$ [M1] $v = 10.5$ m s^{-1} [A1]
		(ii)	After d = 50 m, the car is at maximum speed. The power of the car at that point is $P = Fv = (1000)(10.54)$ [M1] $= 1100$ W [A1] Ecf allowed
	(c)		Since the maximum speed is unchanged but the mass has increased, the final kinetic energy of the car is increased. Since the power of the engine is the same, more time is now needed to for the car to gain this larger amount of kinetic energy.